embedUR

Linux Assingment-4

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For the attached file, write a bash script which should take the file as input and have to go through it line by line and need to generate an output file (say output.txt) with listings of following parameters fetched from the input file.

Params expected to be fetched from input.txt file : **"frame.time", "wlan.fc.type", "wlan.fc.subtype"**

Sample output expected in output.txt:

"frame.time": "Nov 14, 2024 11:44:48.219200000 India Standard Time",
"wlan.fc.type": "1",
"wlan.fc.subtype": "9"
"frame.time": "Nov 14, 2024 11:44:48.219208000 India Standard Time",
"wlan.fc.type": "0",
"wlan.fc.subtype": "1",

and so on.

Command:

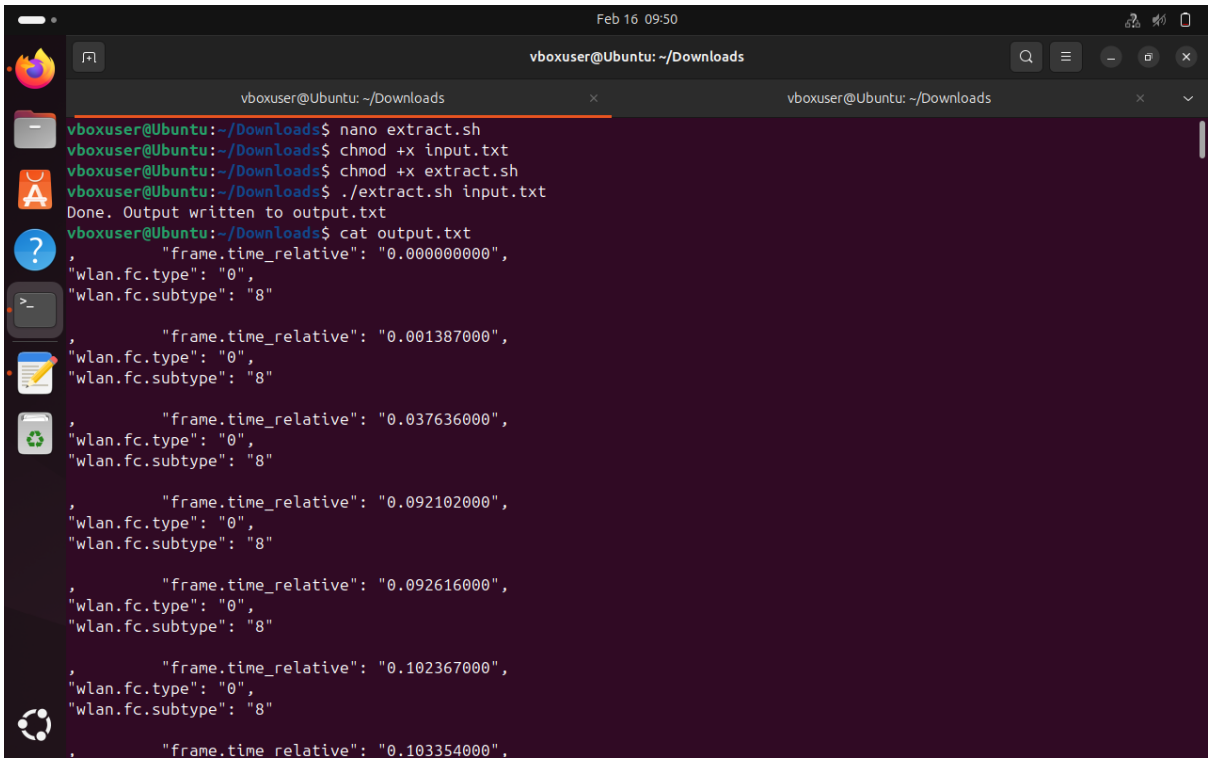
```
cd Downloads
nano extract.sh
chmod +x input.txt
chmod +x extract.sh
./extract.sh input.txt
cat output.txt
```

code:

```
#!/bin/bash

if [ $# -ne 1 ]; then

   echo "Usage: $0 input.txt"

   exit 1

fi


input="$1"

output="output.txt"
```

```
> "$output"

frame=""
type=""
subtype=""

while IFS= read -r line
do
  # match each parameter
  if [[ $line == *"frame.time"* ]]; then
    frame=$(echo "$line" | sed -E 's/.*"(frame\.time)"[: ]*"(.*)".*/"\1": "\2"/')
  fi

  if [[ $line == *"wlan.fc.type"* ]]; then
    type=$(echo "$line" | sed -E 's/.*"(wlan\.fc\.type)"[: ]*"(.*)".*/"\1": "\2"/')
  fi

  if [[ $line == *"wlan.fc.subtype"* ]]; then
    subtype=$(echo "$line" | sed -E 's/.*"(wlan\.fc\.subtype)"[: ]*"(.*)".*/"\1": "\2"/')
  fi

  # once all 3 found → print and reset
  if [[ -n $frame && -n $type && -n $subtype ]]; then
    echo "$frame,"   >> "$output"
    echo "$type,"    >> "$output"
    echo "$subtype"  >> "$output"
    echo             >> "$output"

    frame=""
    type=""
    subtype=""
```

```
  fi

done < "$input"

echo "Done. Output written to $output"
```



Feb 15 17:59

vboxuser@Ubuntu: ~/source

GNU nano 7.2 backup_manager.sh

```bash
#!/bin/bash
shopt -s nullglob
SOURCE_DIR="$1"
BACKUP_DIR="$2"
EXT="$3"
export BACKUP_COUNT=0

if [ ! -d "$SOURCE_DIR" ]; then
    echo "Source directory does not exist"
    exit 1
fi

if [ ! -d "$BACKUP_DIR" ]; then
    mkdir "$BACKUP_DIR"
    if [ $? -ne 0 ]; then
        echo "Failed to create backup directory"
        exit 1
    fi
fi

FILES=("$SOURCE_DIR"/*."$EXT")

if [ ! -e "${FILES[0]}" ]; then
    echo "No files with .$EXT found"
    exit 0
fi

echo "Files to be backed up:"
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy

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vboxuser@Ubuntu: ~/source

GNU nano 7.2 backup_manager.sh

```bash
echo "Files to be backed up:"
for file in "${FILES[@]}"; do
    echo "$(basename "$file") - $(stat -c %s "$file") bytes"
done

for file in "${FILES[@]}"; do
    filename=$(basename "$file")

    if [ -f "$BACKUP_DIR/$filename" ]; then
        if [ "$file" -nt "$BACKUP_DIR/$filename" ]; then
            cp "$file" "$BACKUP_DIR/"
            BACKUP_COUNT=$((BACKUP_COUNT + 1))
        fi
    else
        cp "$file" "$BACKUP_DIR/"
        BACKUP_COUNT=$((BACKUP_COUNT + 1))
    fi
done

REPORT="$BACKUP_DIR/backup_report.log"

echo "Backup Summary" > "$REPORT"
echo "----------------" >> "$REPORT"
echo "Total files backed up: $BACKUP_COUNT" >> "$REPORT"
echo "Backup directory: $BACKUP_DIR" >> "$REPORT"
```

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